Predicting Human Productivity Using Machine Learning Models

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***Abstract*—Productivity is an important factor in our daily life. Human beings want to utilize every moment of their life. Predicting human productivity through their lifestyle and behavior is a game-changing concept. It will benefit workplace efficiency monitoring, increasing personal development and also guide students to monitor their productivity. Employers can identify gaps in their company services. Machine learning models make our task easier. The data set was cleaned and modified with necessary preprocessing. New features were engineered to capture hidden relationships. After processing, multiple models were evaluated. Such as Linear regression, XGBoost, LightGBM, Random Forest, Multi Layer Perceptron. We measured R^2, RMSE, MAE and among these models Random Forest and LightGBM performed best with 0.924 R^2 value. Others performed between 0.89 to 0.91. Fairness analysis was also checked in different groups like gender, age and jobtype. There was constant variation in error rates in subgroups which indicates a potential bias in prediction. The findings highlight that human behaviors have a huge impact on productivity and also demonstrated fairness-aware approaches in different types of subgroups.**

I. Introduction

Social media has become a part and parcel of our life- occupying most of our focus in communication, work or gathering information. However, its constant companionship makes one wonder, for both individual point of view and organization, how regular usage of social media influences productivity? This paper tries to address that with a data-driven study on productivity that embodies human lifestyle focusing on the patterns of social media consumption. The target variable in our study is the continuous `actual_productivity_score` from the dataset, which ensures that we frame the problem in alignment with supervised regression problem rather than a binary classification problem.

We start by pruning the missing target values in the data to uphold the label integrity, while the other gaps are filled with median and most-frequent values for numerical and categorical features respectively. This keeps the sample size as it is and doesn't distort the distributions. Boolean features like `has_digital_wellbeing_enabled` and `uses_focus_apps` are expressed in numerical forms and the categorical features like `job_type`, `gender` or `social_platform_preference` are label-encoded to make them usable in our study. We flag the extreme values by using interquartile-range so that they don't unduly influence the statistical evaluation.

We take new interpretable features such as `social_time_per_work_hour` (a ratio between the work hours and social media consumption hours), `sleep_deficit` (deviation from approximate middle-ground of 8 hours) and `stress_burnout_interaction` (chronic strain tracker) to reflect behavioral mechanisms that influence productivity more accurately. Such derived features translate raw data into something in alignment with how productivity is viewed.

Modeling is initiated by grouping regressors to work on both linear and non-linear structure such as a Linear Regression model, tree-based ensembles models (Random Forest, XGBoost, Light GBM) and also a MLP (Multi-Layer Perception) with feature scaling. Standard regression metrics (R^2, RMSE, MAE) are chosen for summarizing performance by applying hold-out split and cross-validation to nullify the wrong estimations. The emphasis is done not only in predictive performance but also in explanation and fairness in the study. In the study we compute Random Forest feature importances, cover permutation importance and also through SHAP (TreeExplainer) we execute summary plots to keep the features relevant. A function named `subgroup_metrics` is used in computing RMSE/MAE by creating groups such as gender, job type and age group, with dictionaries mapped in interpreting encoded values. We visualize graphs throughout the study to support our interpretations.

All in all, the study exhibits a thorough, interpretable model training crafted to productivity prediction and discussion, focusing on engineered analysis, various model families and post-hoc fairness checks.

II. Literature Review

The paper by Bashiri and Kowsari (2024) [7] demonstrates how Large Language Models (LLMs) and AI tools are transforming student engagement on educational social media platforms. The authors show that integrating AI into these platforms improves academic performance, critical thinking and delivering personalized learning experiences. AI analyzes students' behavior to recommend relevant materials to study groups based on shared skills, and provides instant feedback on assignments which enhances learning speed. It also uses quizzes and badges to maintain motivation and engagement. For educators, AI offers tools for forecasting resource needs. It also manages student support and improves learning outcomes through data analysis. Mental health support is another key benefit. AI detects signs of stress or disengagement and offers timely interventions. The study highlights the role of AI-generated visualizations in simplifying complex concepts, making academic communication more engaging. Using time series analysis (ARIMA) on platform usage data from 2011 to 2024, the authors find that entertainment focused social media use is declining while educational use is rising and they predict that this trend will continue through 2030. To sum up, the paper concludes that LLM and AI integration makes educational social media more personalized, collaborative, and supportive. It benefits both students and institutions.

Nti et al. (2021) [8] studied how students' social media use affects their academic performance using machine learning. The researchers collected survey data from 550 university students, including demographics, time spent on social media, use during lectures, and online activity types. Academic performance was measured by GPA. Two models, Decision Tree and Random Forest, were trained on features such as daily social media hours and lecture-time usage. Results showed that both the frequency and timing of social media use had a noticeable effect on GPA. Students who used social media often during lectures tended to have lower GPAs, suggesting that multitasking can reduce focus. Some students still performed well despite heavy use, showing that study habits and discipline also matter. Between the models, Random Forest gave better predictions due to its ensemble approach. The study highlights how machine learning can help schools identify at-risk students early using behavioral data. Ramzan et al. (2025) [9] examined how social media habits of students relate to their academic performance. Data was collected from 550 students across five departments, with GPA as the target variable. After cleaning incomplete responses, missing values were filled, and categorical data such as department and usage purpose were one-hot encoded. Continuous features like daily usage hours were standardized. For the LSTM model, the dataset was reshaped into time sequences using a sliding window. Three models were tested such as, Decision Tree, Random Forest, and LSTM. The Decision Tree was limited to depth 4, Random Forest used 100 trees, and the LSTM had 128 hidden units with dropout to reduce overfitting. Performance was measured using accuracy, RMSE, MAE, MAPE, and correlation. Results showed that LSTM performed best (81.2

Sharma et al. [10] [2018] examines the relationship between social media usage and work-life balance among 60 corporate employees in Bengaluru, India, focusing on gender differences and levels of internet addiction. Using correlation analysis, Mann-Whitney U, and Kruskal-Wallis tests, the research found a significant negative correlation between internet addiction and two dimensions of work-life balance: intrusion of personal life into work and intrusion of work into personal life indicating that higher online engagement leads to greater interference in both domains. While no gender differences emerged for internet addiction, work enhancement by personal life, or intrusion of personal life into work, differences were found for intrusion of work into personal life and personal life enhancement by work. The findings suggest that excessive social networking blurs boundaries between professional and personal spheres, impacting balance, though moderate use can enhance certain aspects of life. The study recommends strategies for organizations to manage employees' online activity to foster healthier work-life integration.

Ayyash et al. [11] [2022] examines the impact of social media on employee productivity in the workplace, focusing on both its advantages and drawbacks. It highlights that social media can enhance communication, collaboration, knowledge sharing, and professional networking, which may improve creativity and efficiency. However, it also identifies significant risks, including time wastage, distraction, reduced concentration, data security threats, and potential damage to organizational reputation. The study emphasizes that while the effects can be both positive and negative, banning social media outright is impractical in modern workplaces. Instead, it recommends that organizations implement clear policies, guidelines, and monitoring systems to ensure that social media is used productively, leveraging its benefits while minimizing its harmful impacts.

Alzaabi et al. [12] explores how excessive social media use, including addiction and behaviors like "phubbing" (ignoring others to focus on a phone), negatively impacts employee performance, mental health, and workplace culture. Drawing on recent literature, it shows that platforms such as Facebook, Instagram, TikTok, and X can reduce productivity, increase stress, anxiety, and depression, and damage teamwork, communication, and morale. Constant connectivity and fear of missing out (FoMO) contribute to distraction, time mismanagement, and lower work quality. The study also notes organizational consequences like weaker engagement, innovation loss, and a toxic work environment. While acknowledging some benefits of social media, it emphasizes the need for policies, awareness programs, and balanced usage strategies to mitigate harm and maintain a healthy, productive workplace

III. Methodology

A. Data Description

For the dataset we select Kaggle platform as our source. Here the social media vs productivity dataset [1] had 30 thousand values and 19 feature columns initially. Later we added 3 extra columns through feature engineering. We targeted the actual productivity score column. The dataset is only 4 months old as of September 2025. It contains categorical and numerical values. We handled these values according to our needs. A large amount of data helps to train the models in a better way.

B. Data Processing

We did some pre processing in the dataset. At first the dataset contained around 30 thousand data and 19 columns. We found some null values in columns like sleep hour, actual productivity score, stress level, job satisfaction. We removed the rows missing target values. For missing numerical values we filled them with median. We also handled missing categorical values with the most frequent ones. For boolean values we assigned 0 for false and 1 for true. We also had some categorical text columns like gender, job type, social platform preference. We changed them using label encoding. We have used the IQR method to detect outliers in numeric features. Then instead of dropping the rows we capped the outliers. We also checked the correlation of our dataset. *social_time_per_work_hour*, sleep deficit, stress $\times$ burnout interaction).

	Outlier_Count	Outlier_%
has_digital_wellbeing_enabled	6828	24.707798
daily_social_media_time	301	1.089198
number_of_notifications	244	0.882938
screen_time_before_sleep	184	0.665822
coffee_consumption_per_day	118	0.426995
weekly_offline_hours	106	0.383572
work_hours_per_day	88	0.318437
age	0	0.000000
gender	0	0.000000
job_type	0	0.000000
sleep_hours	0	0.000000
stress_level	0	0.000000
perceived_productivity_score	0	0.000000
social_platform_preference	0	0.000000
uses_focus_apps	0	0.000000
breaks_during_work	0	0.000000
days_feeling_burnout_per_month	0	0.000000
job_satisfaction_score	0	0.000000

Fig. 1. Outlier detection summary showing the count and percentage of outliers across all features.

1) ***Outlier Detection Results****: Observation:* The feature `has_digital_wellbeing_enabled` shows the highest proportion of outliers (24.7%), while several features such as `age`, `gender`, and `job_satisfaction_score` contain no detected outliers.

C. Learning Phase

1) Random Forest: Random Forest [2] is an ensemble learning algorithm that is a collection of decision trees which improves accuracy and stability. Each tree is trained on a random subset of the data and thus introduces diversity that overfitting. Random Forest is particularly effective in problems involving mixed data types, such as productivity prediction where demographic, behavioral, and lifestyle features interact in complex ways. It balances predictive accuracy with interpretability.

2) Light Gradient Boosting Machine (LightGBM): LightGBM [3] is a gradient boosting framework developed by Microsoft that increases efficiency, speed, and scalability. Unlike traditional boosting algorithms, LightGBM uses histogram based techniques and grows trees leaf wise instead of level wise. So, it can capture complex patterns while reducing memory usage. It is particularly suited for large datasets with high dimensional features.

3) Extreme Gradient Boosting (XGBoost): XGBoost [4] is an advanced implementation of gradient boosted decision trees designed for efficiency and scalability. Each new tree corrects previous ensemble errors and it builds models in a sequence. XGBoost offers regularization techniques, shrinkage, and handles missing values better than traditional boosting methods. It can also manage high dimensional data and capture nonlinear relationships. In productivity prediction, it can effectively learn from interactions between behavioral features, such as stress, sleep, and work habits.

4) Linear Regression: Linear Regression [5] is one of the most widely used statistical techniques for predictive modeling. It estimates the relationship between a dependent variable and one or more independent variables by fitting a straight line through the data. It is easy to interpret and implement. In our data it allows us to measure the influence of lifestyle and behavioral variables such as sleep, stress, or social media on productivity outcomes.

5) Multi Layer Perceptron (MLP): The Multi Layer Perceptron (MLP) [6] is a type of feedforward neural network consisting of input, hidden, and output layers. Each neuron applies a weighted sum of inputs followed by a nonlinear activation function. It can handle complex nonlinear relations. In productivity prediction, MLP can learn complicated dependencies between behavioral and demographic features.

IV. Results and Discussion

A. Feature Importance

1) ***Correlation signals (w.r.t. actual_productivity_score):***

- **perceived_productivity_score** shows an extremely strong positive correlation ($r = 0.939$), indicating self-perceived productivity almost tracks the target.
- **job_satisfaction_score** is also strongly and positively correlated ($r = 0.842$).
- All other variables exhibit very weak marginal correlations ($|r| \leq 0.011$), e.g., *daily_social_media_time* ($r = -0.010$), *sleep_hours* ($r = -0.008$), *work_hours_per_day* ($r = -0.0001$).

Interpretation: Two attitudinal variables dominate the linear relationship with the target. The behavioral/usage variables add little signal on their own.

2) ***Model-based importance (Random Forest, baseline):*** Top importances from the baseline Random Forest model:

- **perceived_productivity_score** — 0.890

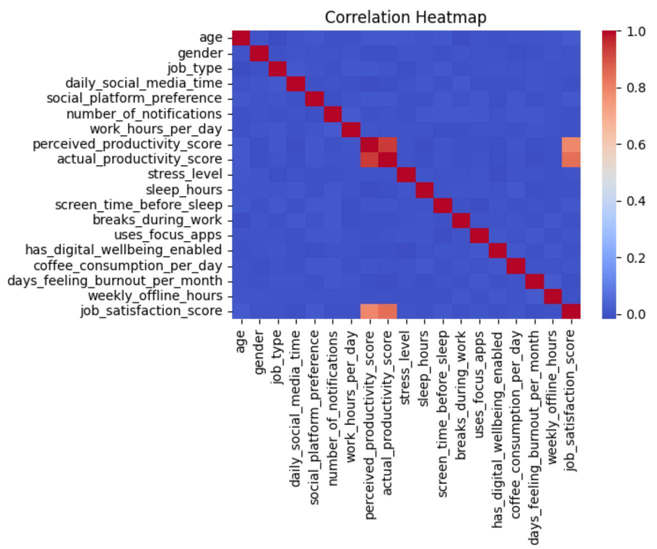

Fig. 2. Correlation Heatmap of features with respect to `actual_productivity_score`.

- **job_satisfaction_score** — 0.052
- *work_hours_per_day* — 0.006
- *sleep_hours* — 0.006
- *weekly_offline_hours* — 0.005
- *screen_time_before_sleep* — 0.005
- *daily_social_media_time* — 0.005

Remaining features each $<$ 0.005 (e.g., age, number_of_notifications, days_feeling_burnout_per_month, stress_level, categorical encodings, etc.).

Interpretation: The model relies overwhelmingly on **perceived_productivity_score**, with **job_satisfaction_score** a distant second. The rest form a long tail with small, roughly comparable contributions.

*3) **Engineered features**:* Engineered variables and their importances in the enhanced Random Forest:

- *stress_burnout_interaction* — 0.00424
- *social_time_per_work_hour* — 0.00378
- *sleep_deficit* — 0.00292

Interpretation: These features are directionally sensible (e.g., chronic stress × burnout days, social time normalized by work hours, sleep deficit from 8h "ideal"), but they add modest incremental signal compared to the top two attitudinal predictors. Their presence did not materially change the dominance pattern.

*4) **SHAP global explanations**:* SHAP confirms the global picture:

- **perceived_productivity_score** has the largest absolute SHAP values; higher values strongly increase predicted actual productivity.
- **job_satisfaction_score** is the next most influential and generally positive: higher satisfaction pushes predictions upward.
- Features like *age*, *daily_social_media_time*, *coffee_consumption_per_day*, and *work_hours_per_day* show small effects with mixed directionality, consistent with their low importances.
- Most other variables have SHAP magnitudes clustered near zero, indicating minimal contribution to the model's decisions.

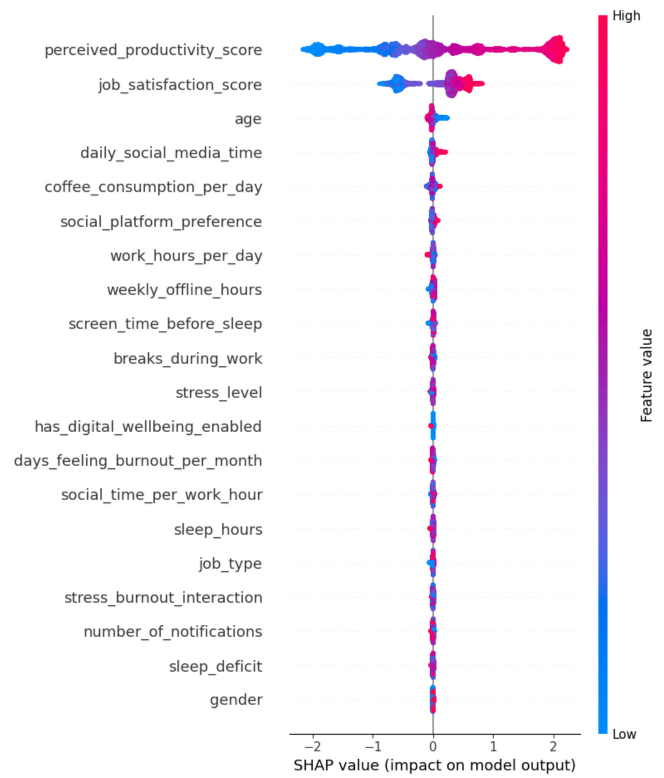

Fig. 3. SHAP summary plot showing feature importance and the effect of feature values on model output.

*5) **SHAP-based feature importance**:* *Interpretation:* The SHAP plot highlights that `perceived_productivity_score` and `job_satisfaction_score` dominate the model's predictions. Other behavioral features such as `daily_social_media_time` and `coffee_consumption_per_day` contribute less but still provide marginal signals.

Interpretation: Both tree-based importance and SHAP agree: predictions are driven mainly by perceived productivity and job satisfaction; everything else fine-tunes.

Caveat: Given the very high correlation ($r = 0.939$) and near-monopoly of importance, **perceived_productivity_score** may act as a near-proxy for the target. This is not "leakage" if both are measured independently, but it does limit actionability. Consider reporting two models in an appendix—with and without **perceived_productivity_score** (and optionally **job_satisfaction_score**)—to reveal the predictive power of purely behavioral features.

B. Performance Metrics

TABLE I
MODEL PERFORMANCE COMPARISON

Model	R^2	RMSE	MAE
Linear Regression	0.905752	0.582695	0.428623
Random Forest	0.924462	0.521661	0.401924
XGBoost	0.914157	0.556105	0.429498
LightGBM	0.924686	0.520886	0.401280
MLP	0.906128	0.581530	0.441874

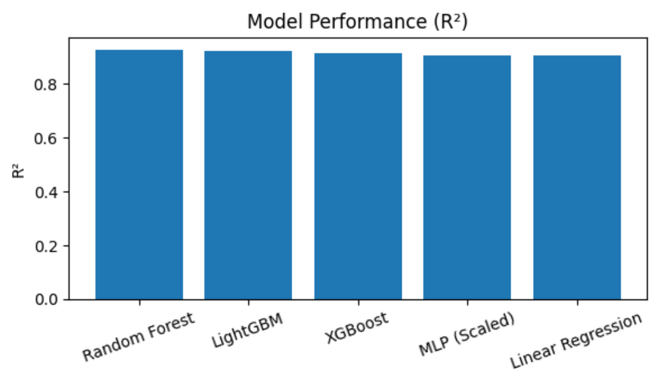

Fig. 4. Comparison of model performance on the hold-out test set using R^2 metric.

1) *Hold-out test results: Ranking & takeaways:*

- **LightGBM** achieved the best scores overall (R^2 0.9247, RMSE 0.5209, MAE 0.4013), narrowly edging out Random Forest.
- The deltas vs. RF are tiny (ΔR^2 0.00022; $\Delta RMSE$ 0.00078; ΔMAE 0.00064), so the two tree ensembles are effectively tied.
- Relative to Linear Regression, LightGBM reduces RMSE by 10.6% (0.5827 → 0.5209) and raises R^2 by 1.9 p.p. (0.9058 → 0.9247), indicating non-linear interactions captured by trees matter beyond simple linear effects.
- XGBoost performs between Linear Regression and the top two, consistent with slightly different default regularization and depth behavior under this feature mix.

Error magnitude: An MAE of 0.40 in the target's native units indicates the model's typical absolute error is less than half a point (on the same scale used to record productivity). Given the strong attitudinal driver, this is a high-fit regime.

2) *Group-level error (fairness diagnostic):*

- **Gender (RMSE):** Female 0.5069, Male 0.5349, Other 0.5330 → spread 0.028 (5% of overall RMSE).
- **Job type (RMSE):** range 0.5075–0.5422 → spread 0.035.
- **Age group (RMSE):** range 0.5056–0.5395 → spread 0.034.

Interpretation: No large performance gaps appear across the reported groups; observed spreads are modest relative to the overall RMSE (0.521). Slightly higher errors for Doctors and the 60+ group may reflect smaller sample sizes or different feature–outcome relationships and are worth monitoring.

3) *Discussion & implications:*

- **Predictive drivers:** Performance is excellent primarily because the model can lean on **perceived_productivity_score** (and to a lesser extent **job_satisfaction_score**). This yields high R^2 but reduces interpretability for intervention: improving "perception" or "satisfaction" is a broad objective, whereas behavioral levers (sleep, breaks, notifications) show small direct effects in this dataset.
- **Actionability check:** For managerial or wellness insights, report an additional model excluding perceived and satisfaction scores; rank the remaining features there. That version will likely surface variables such as *work_hours_per_day*, *weekly_offline_hours*, *screen_time_before_sleep*, and the engineered *stress_burnout_interaction* more clearly.
- **Encoding & outliers:** Categorical variables were numerically encoded. Tree models are robust, but avoid applying outlier treatment designed for continuous variables to binary features in future iterations (e.g., *has_digital_wellbeing_enabled*), since "outliers" don't apply to 0/1 data.
- **Model choice:** Given virtually identical accuracy, choose LightGBM for its speed and tunability, or Random Forest for simplicity and stability. Either is appropriate as the primary production model.

4) *Bottom line:*

- **Feature importance:** Dominated by **perceived_productivity_score**, with **job_satisfaction_score** a distant second; all other features contribute marginally. SHAP and RF importances are fully consistent. Engineered features add small but sensible signals.
- **Performance: LightGBM Random Forest** lead (R^2 0.925, RMSE 0.521, MAE 0.401). Errors are fairly uniform across demographic and job segments.
- **Recommendation:** Present two views—(A) best predictive model (LightGBM) and (B) an "actionability model" without attitudinal targets—to balance accuracy with interpretability for decision-making.

V. CONCLUSION

Conclusion The paper executes a complete roadmap for predicting `actual_productivity_score` that is based on both prediction models with proper interpretation and fair diagnostics. Pragmatic preprocessing is used in data cleaning and domain-informed feature extraction is done to enrich the discussion. Here, multiple models are used in training which are evaluated and compared with each other through R^2, RMSE and MAE while Random Forest features importances, permutation importance, and SHAP enables us to have layered explanations of those predictions. Key conclusions from this study is that the emphasis was done heavily on engineered behavioral features, the proper usage of tree-based regressors and the fairness in the discussions beyond accurate results. However, although the 80/20 data split and its limited tuning and cross-validation shows expected results, with more rigorous CV, random data or with external cohorts, such results could very much change. Dataset biases like sampling, missingness or label reliability might have its effects in predictions and fairness.

VI. Future Work

Keeping future works in mind, the study should include automatic optimization of the models' settings for better performances and also cross-validation testing method to get a more realistic estimation of accuracy. It would be better to have a proper analysis on which features influence more in predicting productivity versus the ones that have lesser influence. To top it all, if the model still shows dissatisfactory outcomes in certain subgroup study, more advanced methods should be applied to get fairer results.

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